

Himanshu Rabha

Assam, India

+91-7002528476

himangshurabha10@gmail.com

LinkedIn

GitHub

EDUCATION

Tezpur University

M.Tech - Electronic Design and Technology

2025 – Present

Tezpur, Assam

Central Institute of Technology Kokrajhar

B.Tech - Instrumentation Engineering

2021 – 2025

Kokrajhar, Assam

RELEVANT COURSEWORK

- Physics and Technology of Semiconductor Devices
- Advanced Semiconductor Devices
- Advanced Logic Circuits
- Mixed Signal Design
- Modeling and Simulation of Digital Systems
- Communication System Design
- Electronic Product Design
- Programming for Electronics Systems

VLSI PROJECTS

UART Transmitter and Receiver | Verilog HDL, FSM Design

2026

- Designed UART transmitter and receiver modules using Verilog HDL.
- Implemented FSM-based communication architecture.
- Developed testbench and verified functionality using simulation waveforms.

Synchronous FIFO Memory Design | Verilog HDL

2026

- Designed synchronous FIFO architecture with read/write logic.
- Implemented status flag generation and buffer management.
- Verified overflow and underflow conditions through simulation.

8-bit Arithmetic Logic Unit (ALU) | RTL Design

2026

- Designed arithmetic and logical operation blocks using Verilog HDL.
- Implemented addition, subtraction, logical and shift operations.
- Validated functionality using testbench-driven verification.

Traffic Light Controller using FSM | Verilog HDL

2026

- Designed FSM-based traffic controller with timing sequence management.
- Implemented state transition logic and signal generation.
- Verified behavior using simulation waveforms.

ACADEMIC PROJECTS

Solar PV Based Cooling System with MPPT Controller | MATLAB, Simulink

2025

- Designed and simulated a solar-powered cooling system using MPPT techniques.
- Analyzed system efficiency under varying environmental conditions.

Speed Control of DC Motor using DC-DC Converter | Control Systems

2024

- Developed converter-based speed control system for DC motor applications.
- Studied speed regulation and dynamic performance through simulation.

INDUSTRIAL TRAINING

Numaligarh Refinery Limited (NRL)

Industrial Training

1 Month

Assam

- Observed industrial instrumentation and process control systems.
- Studied sensors, transmitters, control panels and automation practices.
- Gained exposure to refinery plant operations and maintenance activities.

TECHNICAL SKILLS

HDLs: VHDL, Verilog HDL, SystemVerilog (Basic)

Digital Design: RTL Design, FSM Design, Combinational Logic, Sequential Logic, Timing Concepts

Semiconductor: Semiconductor Device Physics, CMOS Fundamentals, Mixed Signal Design

Tools: ModelSim, Xilinx ISE, Vivado (Basic), MATLAB, Simulink, LTSpice, Multisim, EasyEDA

Programming: C, Python

AREAS OF INTEREST

VLSI Design • RTL Design • Design Verification • ASIC Design Flow

FPGA Design • Semiconductor Devices • Digital System Design